

# Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

**Benchmark:** HOPE\_DAC | **Context:** Global NLP/AI Research Community — Computational Mental Health (HOPE Benchmark Assessment)

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The 12-label DA taxonomy [Q2, Q14, Q17] was developed in consultation with therapist and counseling experts [Q27], explicitly distinguished from general schemes (Switchboard) on domain specificity grounds [Q27], and organized hierarchically [Q29, Q32]. For the elicited use case, the elicitation (A2) affirms that 'the dialogue-act taxonomy is designed to capture intent and conversational flow in a way that is generally applicable across settings' and treats the taxonomy as useful for guiding chatbot dialogue trajectory. However, three structural concerns remain: (1) Single-label assignment is acknowledged by the authors to be a simplification — 'sometimes, an utterance can have multiple dialog-acts' [Q30] — and the elicitation (A3) confirms this is a real concern for downstream chatbot use; (2) The authors themselves note that speaker-initiative/responsive label pairs 'seem complementary in nature' [Q51] with confirmed systematic confusion in error analysis (OD:ID 43%, ID:ACK 28%, YNQ:IRQ 26%) [Q91, Q92]; (3) The taxonomy is grounded in US clinical practice [Q98] and may systematically misclassify communicative acts prevalent in target downstream communities (call-and-response → potentially miscoded as ACK or GC; personalismo opening → possibly GT/GC instead of therapeutically functional rapport-building; spiritual coping → possibly OD or GC). No multi-label counseling DA benchmark currently exists [WEB-15, WEB-18], and no culturally adapted DA taxonomy has been identified [WEB-23], so HOPE is not exceptional but the gap is real.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2     | "We identify the requirement of such conversation and propose twelve domain-specific dialogue-act (DAC) labels." (p.1)                                                                                                                                                                                                                                                                                                   |
| Q14    | "Each utterance in the dataset is tagged with one of the 12 counseling-aligned dialogue-act labels (c.f. Section 3)." (p.2)                                                                                                                                                                                                                                                                                              |
| Q17    | "We propose twelve dialogue-act labels that are aligned with mental-health counseling session." (p.2)                                                                                                                                                                                                                                                                                                                    |
| Q27    | "Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3) |
| Q29    | "Each utterance in the dialogue belongs to one of the three categories – speaker initiative, speaker responsive, and general." (p.3)                                                                                                                                                                                                                                                                                     |
| Q30    | "Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance." (p.3)                                                                                                                                                                                                                          |
| Q32    | "Our annotation scheme assigns three distinct dialogue-act labels to the first two categories, while the remaining four labels belong to the general category." (p.4)                                                                                                                                                                                                                                                    |
| Q51    | "The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature." (p.4)                                                                                                                                                                                                                                                     |
| Q91    | "We observe three pairs with significant error-rates ( $\geq 25\%$ ) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%)." (p.7)                                                                                                                                                                                                                                                                                                  |
| Q92    | "For the prediction of information delivery (ID), SPARTA is confused most of the time with other classes – 19% with PA, 20% with NA, 43% with OD." (p.7)                                                                                                                                                                                                                                                                 |
| Q98    | "Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States." (p.8)                                                                                                                                                                                                                                         |
| WEB-15 | AnnoMI uses 10 annotators with post-annotation survey, demonstrating richer annotator documentation is achievable ( <a href="https://www.mdpi.com/1999-5903/15/3/110">https://www.mdpi.com/1999-5903/15/3/110</a> )                                                                                                                                                                                                      |

|               |                                                                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-18</b> | Scientific Reports (2025): MultiWD covers multi-label wellness dimensions but for social media, not counseling dialogue acts ( <a href="https://www.nature.com/articles/s41598-025-30873-x">https://www.nature.com/articles/s41598-025-30873-x</a> ) |
| <b>WEB-23</b> | JMIR Mental Health (2025) confirms 'limited attention to linguistic or cultural adaptation' in LLM mental health studies ( <a href="https://mental.jmir.org/2025/1/e78410">https://mental.jmir.org/2025/1/e78410</a> )                               |

## Strengths

- Taxonomy was developed with therapist and counseling expert consultation [Q27], grounding it in domain knowledge.
- Hierarchical organization (initiative/responsive/general) [Q29, Q32] provides a coherent structural framing.
- Explicit differentiation from general-purpose DA schemes [Q27] addresses domain specificity.
- Authors transparently flag the multi-label simplification [Q30] and label-pair complementarity [Q51], enabling informed downstream interpretation.

| ID         | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q27</b> | "Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3) |
| <b>Q29</b> | "Each utterance in the dialogue belongs to one of the three categories – speaker initiative, speaker responsive, and general." (p.3)                                                                                                                                                                                                                                                                                     |
| <b>Q30</b> | "Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance." (p.3)                                                                                                                                                                                                                          |
| <b>Q32</b> | "Our annotation scheme assigns three distinct dialogue-act labels to the first two categories, while the remaining four labels belong to the general category." (p.4)                                                                                                                                                                                                                                                    |
| <b>Q51</b> | "The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature." (p.4)                                                                                                                                                                                                                                                     |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                          |
|-------------|--------------------------------------------------------------|
| <b>OO-1</b> | Review output label categories for regional relevance        |
| <b>OO-2</b> | Identify missing regionally specific categories              |
| <b>OO-3</b> | Flag categories encoding non-regional values                 |
| <b>OO-4</b> | Consider stakeholder-driven taxonomy redesign                |
| <b>OO-5</b> | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** The 12 labels [Q33–Q44] are plausibly relevant to general counseling dialogue. For the elicited individual-counseling-chatbot use case (A2), the taxonomy is treated as generally applicable.

**OO-2:** Missing categories of concern: no labels for spiritually framed coping, narrative-style extended disclosure, indirect disclosure via family narrative, or call-and-response acknowledgment patterns — all flagged in regional context as therapeutically functional in target downstream communities. No published culturally adapted counseling DA taxonomy exists [WEB-23].

**OO-3:** The taxonomy implicitly encodes US clinical assumptions [Q98] — e.g., GC ('General Chit-Chat...possibly because of the vagueness and the lack of sense in the context') [Q44] is a residual category that risks absorbing therapeutically functional culture-specific moves (personalismo rapport-building, spiritual framing) per the regional context.

**OO-4:** Stakeholder-driven taxonomy redesign would be appropriate for culturally adapted deployment, per the elicitation A2 framing this as 'complementary/future work.' No such redesign has been undertaken in the field [WEB-23].

**OO-5:** Documented taxonomy issues: (a) single-label simplification [Q30] mismatches multi-intent reality [A3]; (b) US-clinical grounding [Q98] risks structural misalignment for target downstream communities; (c) confirmed systematic confusion among complementary label pairs [Q51, Q91, Q92]; (d) ACK is a weak point (F1=47.86%) [Q77] with documented confusion with ID.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27    | "Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3) |
| Q30    | "Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance." (p.3)                                                                                                                                                                                                                          |
| Q33    | "Information Request (IRQ): This label is used as a request for some information, e.g., 'Tell me your name.'" (p.4)                                                                                                                                                                                                                                                                                                      |
| Q34    | "Yes/No Question (YNQ): The YNQ label is similar to IRQ; however, the expected response is a trivial yes or no answer." (p.4)                                                                                                                                                                                                                                                                                            |
| Q35    | "Clarification Request (CRQ): This label is assigned to those utterances in which a speaker usually asks the therapist for further clarification about topic that is being currently discussed." (p.4)                                                                                                                                                                                                                   |
| Q36    | "Opinion Request (ORQ): The ORQ label is used when the speaker seeks opinions of the listener." (p.4)                                                                                                                                                                                                                                                                                                                    |
| Q37    | "Clarification Delivery (CD): This label is used when the speaker provides further clarifications about a topic/entity under discussion." (p.4)                                                                                                                                                                                                                                                                          |
| Q38    | "Information Delivery (ID): When the speaker provides some factual objective information about herself." (p.4)                                                                                                                                                                                                                                                                                                           |
| Q39    | "Positive Answer (PA): These labels are used when the utterance is an answer in the form of a simple yes to a question that was previously uttered." (p.4)                                                                                                                                                                                                                                                               |
| Q40    | "Negative Answer(NA): It is used when the utterance is an answer of the form of a simple No to a question asked earlier." (p.4)                                                                                                                                                                                                                                                                                          |
| Q41    | "Opinion Delivery (OD): When the speaker explicitly lists out her/his opinions." (p.4)                                                                                                                                                                                                                                                                                                                                   |
| Q42    | "Greeting (GT): Each session usually starts with greeting by one speaker and an appropriate response from the other." (p.4)                                                                                                                                                                                                                                                                                              |
| Q43    | "Acknowledgment (ACK): In normal conversation, very often, we utter (e.g., 'Yeah! You are right.') to acknowledge the other speaker or to show our agreement without an explicit information request, question, or command." (p.4)                                                                                                                                                                                       |
| Q44    | "General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)                                                                                                                                                                                                             |
| Q51    | "The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature." (p.4)                                                                                                                                                                                                                                                     |
| Q77    | "We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%." (p.7)                                                                                                                                                                                                                               |
| Q91    | "We observe three pairs with significant error-rates ( $\geq 25\%$ ) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%)." (p.7)                                                                                                                                                                                                                                                                                                  |
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## Information Gaps

- Whether expert consultants who helped design the taxonomy [Q27] included clinicians experienced with target downstream communities is not documented.

## Requires Expert Verification

- Cross-cultural counseling experts should review the 12 labels for whether they adequately partition the DA space for target downstream communities, particularly regarding GC, ACK, GT, and ID boundaries.

## Remediation

**Gap 1: Single-label assignment [Q30] mismatches multi-intent utterance reality (elicitation A3) and complementary label pairs show systematic confusion (OD:ID 43%, ID:ACK 28%) [Q91].**

**Recommendation:** Extend HOPE annotations to support primary + secondary labels per utterance on a sample, enabling multi-label and ranked-output evaluation protocols. Report joint top-2 accuracy in addition to top-1 macro-F1 to better reflect downstream chatbot generation needs.

| ID  | Evidence                                                                                                                                                                                        |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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